

Grid-forming converters: Sufficient conditions for RMS modeling

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ABSTRACT

Time-domain simulations are a critical tool for power system operators. Depending on the instability mechanism under consideration and the system characteristics, such as the time constants of controllers, either phasor or Electro-Magnetic Transient (EMT) models should be employed. On the one hand, EMT models provide a detailed modeling of the system dynamics, thus increase the reliability of stability analysis; on the other hand, using these models increase the computing time of simulations, slowing down the security assessment process. To decrease computational time, system operators could resort to phasor-mode simulations for a subset of disturbances. This paper investigates the appropriateness of phasor-approximation models on simulating events related to power supply and balance stability when grid-forming converters are employed for providing frequency support. Moreover, it provides sufficient conditions and bounds for the control parameters of a grid-forming converter under which the phase-approximation modeling of the component is still valid. The control design is carried out using a simple transfer function approach. Time-domain simulation are performed to validate the transfer function analysis using different control settings and different line impedance characteristics.

1. Introduction

The dynamic nature of the electric power system undergoes a period of significant changes. A major part of these changes is associated with the replacement of synchronous generators with converter-based resources, which lead to the so-called low-inertia systems [1–5]. Due to this transition, transmission system operators encounter critical stability challenges for stabilizing the frequency of the system, since in the absence of synchronous generators, the maximum frequency deviation and rate of change of frequency become larger. To preserve a minimum amount of inertia, transmission system operators will rely on new converter control strategies, which can potentially address these issues [6]. These control strategies can be categorized based on the operating mode of the converter [7]. There are two main categories, namely grid-following and grid-forming converter. Considering that grid-following converters rely on a strong grid with high short circuit ratio and penetration of synchronous generation, they cannot be used as a potential control strategy for participating in active power balance [8]. Thus, in this work the focus is on the control strategies for grid-forming converters.

In traditional power systems, the dynamics of transmission lines is usually neglected and an algebraic model of the transmission network is used, i.e. algebraic relationship between the line currents and bus

voltages. This neglecting of the dynamic nature of the network line dynamics is valid since the characteristic time constants of the synchronous generators are much greater. However, the characteristic time-constants of the converter-based resources are much smaller, thus there is no distinct time-scale separation between converter control loops and eigen-frequencies in power network. As a result, the influence that network line dynamics have on system stability is growing and the quasi-static modeling of transmission lines, i.e. RMS modeling [9,10], becomes inappropriate for determining the stability boundaries of the system. Consequently, to analyze fast power components, such as Power Electronic (PE) devices, the electro-magnetic dynamics (network line dynamics) are vital for system stability. However, taking electro-magnetic dynamics into account drastically increases the computational time for solving the system of differential algebraic equations which describes the power system dynamics (since the number of state variables increases).

This creates a limitation for online security assessment, since EMT simulations are considerably slower than RMS simulations. To assess the problem of the computation time of simulation, we first need to understand when RMS modeling is appropriate for simulating zero-inertia systems. As a result, neglecting the line dynamics can lead to incorrect results, such as the RMS model predicting stability while in reality the system is unstable [11]. Several works have focused on parametric

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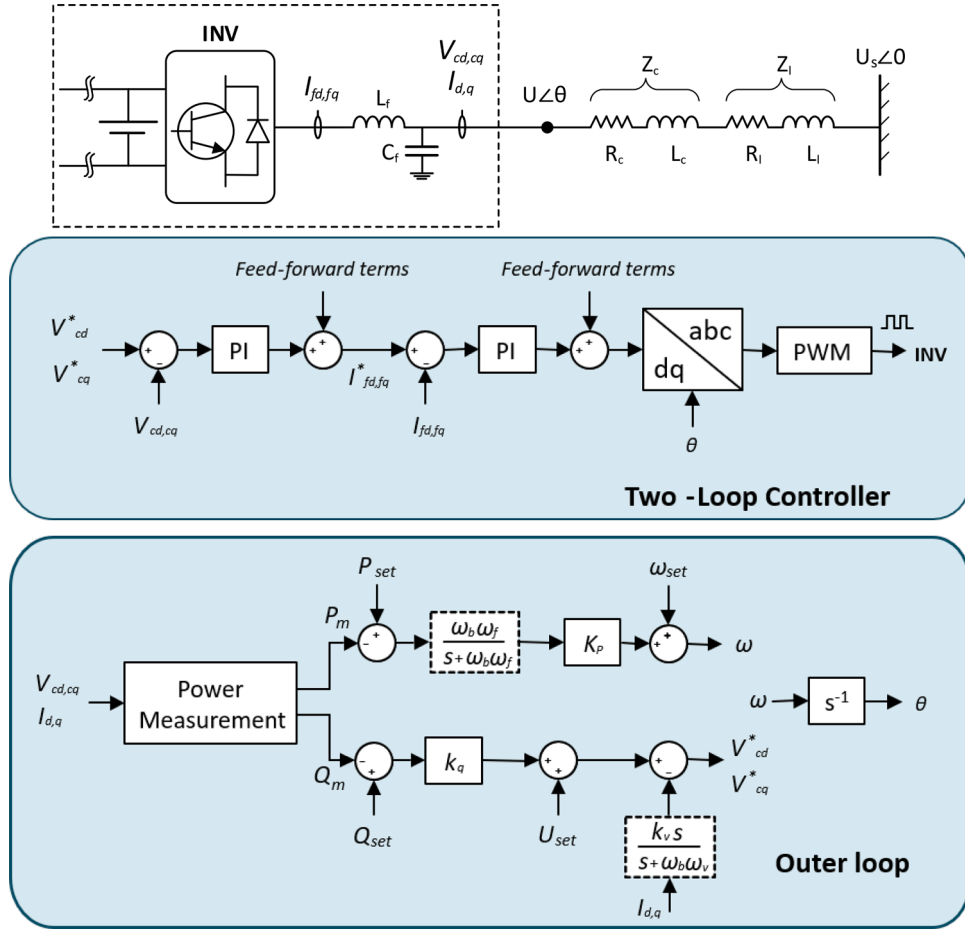


Fig. 1. Control structure of grid forming converter. Dashed blocks are either considered or disregarded depending on the class of the grid forming converter.

constraints for the control settings of grid forming converters considering the influence of network line dynamics on the stability of the converter. While most of the existing work focuses on eigenvalue, bode plot and Nyquist analyses [12–15] for the control tuning of the converter, few researchers have studied the appropriateness of quasi-static modeling of network line dynamics [16–18]. In [16], the authors examined the appropriateness of RMS modeling by means of singular perturbation techniques. In particular, they focused on the timescale of the line and load dynamics in order to neglect the fast dynamics and define a reduced order model which preserves the state variables of the slower dynamics. However, they did not provide any bounds on the control parameters of the outer controllers of the grid-forming converters. Similar approach which only focuses on neglecting the electromagnetic dynamics and fast inner loops based singular perturbation theory have also been presented in [11] and [19]. In [17], the authors use tools both from singular perturbation and Lyapunov theory to derive sufficient conditions on the frequency and voltage droop gains of the converters that guarantee the stability not only of the EMT model but also for the RMS model. However, the bounds for the control gains are quite conservative, due to the employment of Lyapunov theory, which is also highlighted and demonstrated in the results section. It should be also mentioned that the aforementioned works only focus on microgrids where the X/R ratio of the grid impedance is low, which improves significantly the resonance at the nominal frequency. Finally, the authors in [18] propose sufficient conditions, based on transfer function analysis, for a virtual synchronous generator to ensure the stability and the validity of RMS modeling of the grid-forming converter. However, these conditions only focus on the phase crossover frequency of the open loop system and the ratio between the damping and inertia term of the

active power controller of the grid forming converter.

The scope of this paper is to identify the appropriateness of RMS modeling of a class of grid-forming converters, namely power synchronization loop and virtual synchronous generator [7], which are the most commonly used in literature. Our goal is to assess under which conditions RMS modeling (used by simulation tools, such as PowerFactory DigSilent, PSSE, etc.) can still be used for performing power system dynamic security assessment in power grids with high penetration of grid-forming converters. The motivation is that system operators could continue using the RMS for monitoring the active power sharing in systems, as long as the mismatch between the RMS and the EMT models are within acceptable limits. The contributions of this paper are the following:

1. We provide sufficient conditions and upper bounds for the control gains of the outer controllers of grid-forming converters that preserve the time-scale separation properties between the outer control loop and the network line dynamics. We also provide sufficient conditions for the stability of the fully detailed model. To extract the sufficient conditions, transfer function analysis is employed.
2. We present an analytical control tuning of the grid-forming converters, that enable to define the control gains based on minimum phase margin and gain margin requirements for the open loop system of the active power controller.
3. We make use of the gain margin and the phase margin as performance metrics for evaluating the appropriateness of neglecting the effect of the transmission-line dynamics when monitoring the active power balance of the system. This can allow transmission system operators to enforce certain requirements to vendors regarding the

control truing and design of converter operating in grid-forming mode. To this end, we present a transfer function for the relation between the angle and the active power that accounts for the impact of voltage droop on the angle dynamics. This allows us to quantify the impact of the voltage droop on the gain and phase margins of the open-loop system of the active power controller and reducing the coupling between the active and reactive power.

4. We demonstrate the effectiveness of the conditions using two different simulation tools, namely Matlab and PowerFactory. It should be noted that the sufficient conditions are valid for investigating a selection of type of disturbances, namely active power mismatches, grid topology changes and controller interactions and for a short circuit ratio ranging within [1, 3].

The rest of the paper is organized as follows. In Section II, we present a description of the outer loop controller of a grid-forming converter. In section III, we derive sufficient conditions for the control gains of the controllers of the grid-forming converter using transfer function analysis. Section IV verifies the conditions derived in section III using time-domain simulations. Conclusions are drawn in Section VI.

2. Grid-forming converter

As described in [7], grid-forming converters are equipped with a two-loop control system, which is illustrated in Fig. 1. The inner loop is designed to be much faster than the outer loop and the control gains are chosen according to the tuning procedure presented in [20], so that we achieve a time-scale separation between the network line and outer loop dynamics. Similar time-scale separation has been also observed with the inner gains proposed in [13,19]. This enables to study only the impact of the network line dynamics on the outer loop's control gains, that are the ones defining the power regulation in a future low-inertia grid. Therefore, the dynamics presented in this section only concern the outer loops.

2.1. dq and xy frame

Similar to [21], we define two rotating frames: i) the dq rotating frame and ii) the xy rotating frame. The angular speed of the dq frame is equal to the frequency of the i^{th} converter ω_i , whereas the angular speed of the xy frame is equal to the frequency of the grid ω_g . Similar to [22], the converter dq frame leads the xy frame by the angle $\Delta\theta$, which is given by:

$$\frac{d\Delta\theta}{dt} = (\omega_i - \omega_g)\omega_b \quad (1)$$

where $\omega_b = 2\pi f_{nom}$ rad/s is the base angular speed, where f_{nom} is the grid's nominal frequency. The transformation from the xy to dq coordinates is given by:

$$\mathbf{x} = \mathbf{x}^c e^{-j(\theta_0 + \Delta\theta)} \quad (2)$$

where the superscript c denotes the xy frame. The corresponding dq frame is denoted without a superscript, e.g., $\mathbf{x}$. The quantities in dq- and xy frame are described in complex space vector form as:

$$\mathbf{x} = x^d + jx^q \quad (3a)$$

$$\mathbf{x}^c = x^x + jx^y. \quad (3b)$$

2.2. Phase reactor and transformer

As depicted in Fig. 1, the circuit model consists of a Voltage Source Converter (VSC) connected through an RL impedance to the rest of the grid. L_c and L_l represent the transformer and grid inductances, respectively. R_c and R_l represents the transformer and grid resistances, respectively. For the rest of this paper, we consider that $R_g = R_c + R_l$ and

$L_g = L_c + L_l$. The relation between the current flowing through the RL circuit and the voltages U and U_s , expressed in the dq frame, is given by:

$$U = U_s^c e^{-j(\theta_0 + \Delta\theta)} + [R_g + (s + j\omega_i)L_g] \mathbf{i} \quad (4)$$

where L_g and R_g are expressed in p.u. and s is the Laplace complex variable. U_s^c is the grid voltage expressed in the xy frame and U is the converter terminal voltage in the dq frame.

2.3. Voltage-vector control

A voltage set-point is used to control the output voltage U of the converter at its effective terminal bus. An additional voltage droop is employed for regulating the voltage set point based on the reactive power injected. Apart from setting the voltage at the converter terminal, the voltage-vector control unit must also be able to provide damping of poorly-damped resonances. To this end, a high-pass filter current control is incorporated to increase the damping of high frequency modes [23]. To this end, the voltage-vector control can be expressed as:

$$U = U_{set} - \underbrace{\frac{k_v s}{s + \omega_v}}_{H_{hp}(s)} \mathbf{i} + K_q (Q_{set} - Q) \quad (5)$$

where k_v determines the damping effect on the poorly-damped eigen-frequencies. The cut-off frequency of the high-pass filter ($H_{hp}(s)$) is determined by the value of ω_v , which according to [22] takes values in the range $0.1\omega_1 \rightarrow 0.2\omega_1$, where ω_1 is the reference angular speed chosen to be 1.0 p.u. The outer voltage control loop is also equipped with a QV droop characteristic, where K_q is the voltage droop which determines the amount of reactive power change after a voltage deviation at the Point of Common Coupling (PCC).

2.4. Active power control schemes

Two frequency control schemes are considered and described in the following, namely power synchronization loop and virtual synchronous generator.

2.4.1. Power synchronization loop

The active power droop control regulates the angular speed ω_i ; the controller acts by adjusting the voltage angle θ_i based on deviations of the injected/absorbed active power P with respect to a given reference value P_{ref} . The equations describing the imposed frequency and angle at the converter terminal are given by:

$$\omega_i = \omega_1 + K_p (P_{ref} - P) \quad (6a)$$

$$\Delta\theta = (\omega_i - \omega_g) \frac{\omega_b}{s} \quad (6b)$$

where the frequency droop K_p is used to regulate the frequency ω_i based on the power mismatch between the power reference point and the power output.

2.4.2. Virtual synchronous generator

A low pass filter is placed on the active power measurement, as depicted in Fig. 1, that emulates the inertia response of the synchronous generators. The equation describing the imposed frequency and angle at the converter terminal are given by:

$$\omega_i = \frac{\omega_f \omega_b}{s + \omega_f \omega_b} (\omega_1 + K_p (P_{ref} - P)) \quad (7)$$

where ω_f is the cut-off frequency of the low-pass filter. It should be mentioned that this grid-forming scheme is same with the one presented in [18], for $K_p = \frac{1}{D}$ and $\omega_f = \frac{D}{M}$, where D is the damping and M the virtual inertia.

Table 1

Contribution of the active power signal to system modes - Power Synchronization Loop.

y_k [pu]	$ \pi_{k,i} _{\max}$	Mode	x - dom	$ p $
P [pu]	0.50529	$-96.2 \pm j312$	I_d, I_q	0.5052
			I_{fd}, I_{fq}	0.0458
			ξ_d, ξ_q	0.0439
			θ	0.020

Table 2

Contribution of the active power signal to system modes - Virtual Synchronous Generator.

y_k [pu]	$ \pi_{k,i} _{\max}$	Mode	x - dom	$ p $
P [pu]	0.4983	$-102 \pm j315$	I_d, I_q	0.4983
			I_{fd}, I_{fq}	0.0462
			ξ_d, ξ_q	0.0443
			M_{ad}, M_{aq}	0.0146

2.5. Neglecting the inner-loop and filter dynamics

To justify the neglecting of the inner loop and filter dynamics of the converter, we demonstrate the coupling between the system state variables and the active power signal. We focus only on the active power signal, since in this paper we want to evaluate the appropriateness of RMS modeling for monitoring the active power regulation. However, the framework presented in this section can be applied for evaluating the coupling of the state variables of the system to any algebraic variable, e. g. reactive power and voltage signal. We make use of the approach presented in [2]. Using the same notation as in [2], we consider the following set of differential algebraic equations describing the EMT model of a grid-forming converter:

$$\dot{x} = f(x, y) \quad (8)$$

$$0 = g(x, y) \quad (9)$$

where x is a vector matrix containing the system state variables, and y is a vector matrix containing the algebraic variables. To obtain the participation matrix P_x , we first linearize the system around the operating point (x_0, y_0) as follows:

$$\Delta \dot{x} = f_x \Delta x + f_y \Delta y \quad (10)$$

$$0 = g_x \Delta x + g_y \Delta y \quad (11)$$

where f_x , f_y , g_x and g_y are the Jacobian matrices evaluated at the

operating point (x_0, y_0) . The participation factors of the algebraic states, (the active power signal is an algebraic state of the system) can be calculated by the following expression:

$$\Pi_{(y)} = -g_y^{-1} g_x P_x \quad (12)$$

where $\Pi_{(y)}$ is the participation matrix of the algebraic states in system modes, with $\pi_{k,i}$ being the element of the matrix (k denotes the k -th algebraic variable, i denotes the i -th mode). $\pi_{k,i}$ provides information about for the coupling between the algebraic and state variables of the system. Table 1 shows the mode with the highest contribution from the active power signal $|\pi_{k,i}|_{\max}$, as well as the contribution from the state variables of the system $|p|$. The state variables corresponding to the line currents presents the highest contribution to that mode. The state variables (I_{fd}, I_{fq}), (ξ_d, ξ_q) and θ , corresponding to the phase reactor currents, inner voltage control loop and phase angle, respectively, present a small participation to the model. Similar observations, can be made for the virtual synchronous machine, see Table 2. The main difference is that instead of the phase angle, the state variables corresponding to the high pass filter of the virtual impedance controller slightly contribute to the active power signal. Based on this analysis we omit the inner loop, the phase reactor and the filter capacitance dynamics.

Fig. 2 shows a comparison between the full-detailed EMT model and the reduced EMT model (only accounts for the network line dynamics, the outer loops and the virtual impedance controller). For the comparison we consider a line outage, i.e. sudden drop of the short circuit ratio from 5 to 3. As it can be seen, the difference between the EMT models is negligible even for large disturbances, which implies the limited effect of the inner loops and filters on the active power signal.

To summarize, this paper investigates the control tuning of the outer loop and virtual impedance controllers and suggests values for which the bandwidth of the active power controller is limited. This is being done in order to ensure that its effect on the line resonance is minimized and improve the damping ratio of the mode at $s = -\frac{R}{L} \pm j\omega_b$, which is the one that affects significantly the trajectory of the active power signal.

3. Sufficient conditions for RMS modeling

In this section, we identify the control gains and parametric constraints, under which the line dynamics can be disregarded. For the analysis, the system depicted in Fig. 1 is used. The constraints will allow system operators to rely on the RMS modeling of grid-forming converters and not use the EMT model for monitoring the active power sharing in the power grid. For the analysis we use the theory of loop-shaping control design and Nyquist's criterion. In particular, we employ the gain and phase margin g_m and ϕ_m , respectively, which are relative measures of the distance of the Nyquist curve to the critical point $(-1,0)$ [24].

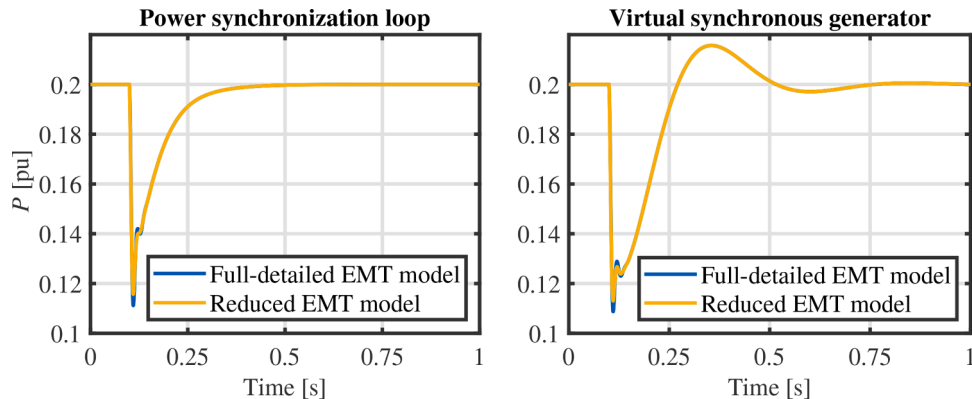


Fig. 2. Trajectory of the active power signal for a step change of the SCR from 5 to 3. The difference between the fully detailed EMT model and the reduced EMT model is that in the latter the inner loop and the filters (phase reactor and capacitance) are disregarded.

To derive the sufficient conditions, we consider the transfer function of the open loop system $L(s)$. The open loop transfer function $L(s)$ consists of the transfer functions $P(s)$ and $C(s)$, which describe the plant and controller dynamics, respectively:

$$L(s) = P(s)C(s), \quad (13)$$

where $C(s) = \frac{K_p}{s}$ in the case of power-synchronization loop and $C(s) = \frac{K_p}{s} \frac{1}{s + \omega_f}$ in the case of the virtual synchronous machine. $P(s) = G_{op}(s)$ is the transfer function of the plant. The transfer function can be found in the appendix. The gain margin is calculated as follows:

$$g_m = \frac{1}{|L(j\omega_{180})|}, \quad (14)$$

where ω_{180} is the phase crossover frequency and is the lowest frequency where the Nyquist plot intersects the negative real axis. The gain margin indicates how much the controller gain can be increased before reaching the stability limit. Similarly, the phase margin is defined as:

$$\phi_m = \pi + \arg L(j\omega_c), \quad (15)$$

where ω_c is the gain crossover frequency where the loop transfer function $L(s)$ has unit magnitude and indicates the amount of phase lag required to reach the stability limit.

To derive the sufficient conditions and provide meaningful expressions, we initially consider that the converter is at no load condition, which yields that i_{q0} and i_{d0} are equal to zero. Later in this section, we will show how the active and reactive currents affect the parametric constraints. Moreover, we will analyze the impact of the control and system parameters on the frequency droop value K_p , as well as on the inertia effect that can be provided by the converter.

From control design perspective, minimizing the differences between the RMS and EMT models of a grid-forming converter can be considered as the problem of minimizing the effect of the network line dynamics on the outer loops of the converters. A reasonable control design requirement is to tune the control gains of the converters in such a way that the bandwidth of the active power controller is less than $0.1\omega_1$ [18,25], where ω_1 is the nominal frequency of the power grid. This yields the following condition for the gain crossover frequency of the open loop system of the active power controller, $\omega_c \leq 0.1\omega_1$. This requirement is due to the line resonance at nominal frequency (modes are at $-\frac{R_g}{L_g} \pm j\omega_1$). By limiting the bandwidth of the controller, we first mitigate the effect of line resonance (increasing the gain margin of the open loop system), but also enable the time-scale separation between the network line dynamics and the outer loop. This ensures that the RMS model of the grid-forming converter is a good approximation of the full-detailed model. However, as it will be presented in Section 3.2, an additional condition is required based on the gain margin of the open loop system in order to improve this approximation.

To this end, we first investigate the properties of the open loop system of the power-synchronization control loop and then the virtual synchronous generator one.

3.1. Power synchronization loop

At no-load condition, the transfer function of the open loop system can be expressed as follows:

$$G_{ol}(s) = \frac{K_p}{C(s)} \frac{(K_q + \omega_1 L_g) U_{set}^2}{\underbrace{K_q \omega_1 L_g + (\omega_1 L_g)^2}_{G_{hp}(s)} + (H_{hp}(s) + R_g + sL_g)^2} \quad (16)$$

We first calculate the phase crossover frequency from setting the imaginary part of the open loop system equal to zero ($\Im(L(j\omega_{180}))=0$).

$$\omega_{180} = \frac{\sqrt{(R_g + k_v)^2 + L_g^2 \omega_1^2 + K_q \omega_1 L_g}}{\omega_1 L_g}. \quad (17)$$

Having calculated the phase crossover frequency, we can now express K_p as a function of the gain margin, network line parameters and the active damping controller parameters. This is given by:

$$K_p = 2(k_v + R_g) \left(\frac{1}{g_m U_{set}^2} + \frac{\frac{(k_v + R_g)^2}{\omega_1 L_g}}{g_m (K_q + \omega_1 L_g) U_{set}^2} \right) \quad (18)$$

where K_p and g_m are inversely proportional to the voltage droop K_q and directly proportional to the damping coefficient k_v . An analytical condition for the stability of the grid-forming converter and the maximum value of the frequency droop is obtained when $g_m = 1$. As described earlier, in order to mitigate the impact of the line resonances, the gain crossover frequency of the open loop system must be less than 0.1 of the nominal frequency ($\omega_c \leq 0.1\omega_1$). This condition allows for a sufficient phase margin close to 90° , since the impact of the pole $-\frac{R_g}{L_g} \pm j\omega_1$ on the phase angle of the open loop system is negligible for $\omega \leq \omega_c$ [24], making the open loop system behaving as an integrator for $\omega \leq \omega_1$. It should be noted that the phase margin of the RMS model is 90° since the open loop system is a static gain (see Appendix). In order to determine the gain crossover frequency, the expression (15) is used with $C(s) = \frac{K_p}{s}$ and $P(s) = G_{op}(s)$. The Laplace complex variable s is set to $j\omega_c$ and $|L(j\omega_c)| = 1$ is solved for ω_c . This yields the following expression for the gain crossover frequency:

$$\omega_c \approx \frac{K_p U_{set}^2}{\omega_1 L_g + \frac{R_g^2}{\omega_1 L_g + K_q}} \quad (19)$$

$$K_p \leq \frac{0.1}{U_{set}} \left(\omega_1 L_g + \frac{R_g^2}{K_q + \omega_1 L_g} \right) \quad (20)$$

It is evident that for small resistances the impact of the voltage droop on the frequency droop value is negligible. Substituting (20) into (18) we obtain the following inequalities (21a)-(21c).

$$k_v \geq \frac{g_{m,min}}{20} L_g - R_g \quad (21a)$$

$$K_q \geq 0 \quad (21b)$$

$$g_m \geq 20 \frac{k_v^3 + 3\beta_2 k_v^2 + \beta_1 k_v + \beta_0}{(\omega_1 L_g)^3 + K_q (\omega_1 L_g)^2 + \omega_1 L_g R_g^2} \quad (21c)$$

where $g_{m,min}$ is a minimum value for the gain margin of the open loop system, which is selected in order to mitigate the line resonance, $\beta_2 = R_g$, $\beta_1 = (K_q \omega_1 L_g + (\omega_1 L_g)^2 + 3R_g^2)$ and $\beta_0 = (K_q \omega_1 L_g R_g + \omega_1 L_g^2 R_g + R_g^3)$. It is worth mentioning that $g_m > g_{m,min}$, due to the fact that the high pass filter is not a static gain. In case $H_{hp}(s) = k_v$, then $g_m = g_{m,min}$. We can notice that by selecting $\omega_c \leq 0.1\omega_1$, the lowest obtainable value of the gain margin is equal to 2 ($g_m \geq 2$), which is equivalent to neglecting the active damping controller. Additionally, from (21a) we can infer that for low X/R ratio the active damping controller $H_{hp}(s)$ can be disregarded, e.g. for $g_{m,min} = 10$ and X/R=1 the lower bound of k_v becomes negative meaning that the mode $-\frac{R_g}{L_g} \pm j\omega_1$ is well damped and not additional damping is required.

3.1.1. Illustrative case

The following example will help us understand better the sufficient conditions (21a)-(21c). It should be noted that the regions shown in Fig. 5 do not represent the stable region of the system, but the region

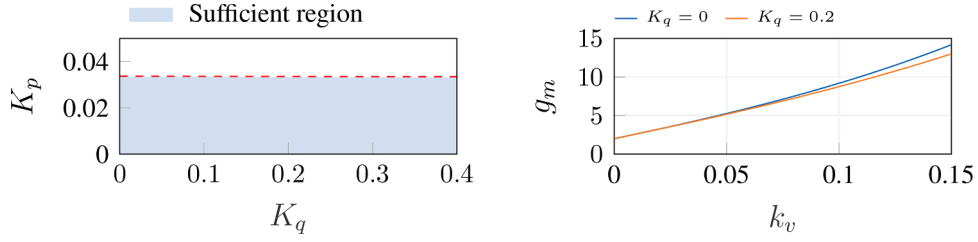


Fig. 3. Left: sufficient region defined by the values of K_p and K_q . Right: impact of damping coefficient k_v and K_q on the gain margin.

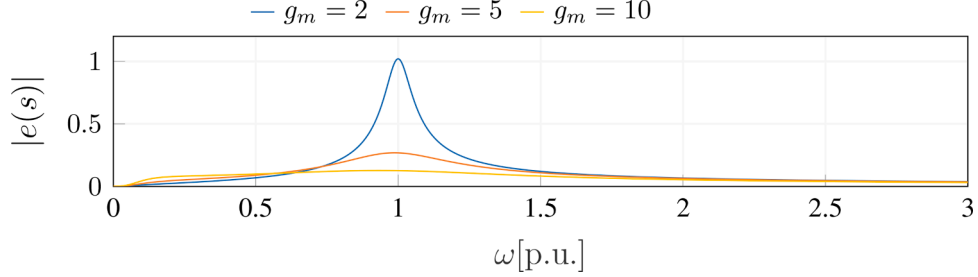


Fig. 4. Singular values of $e(s)$ transfer function for different values of the gain margin.

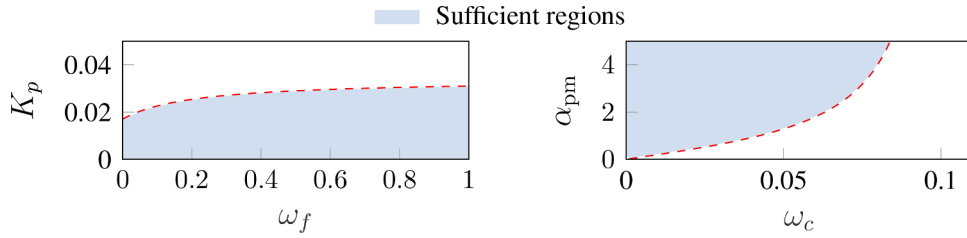


Fig. 5. Sufficient regions for the phase margin and the control parameters of the grid-forming converter.

that establish the timescale separation between the network line and the angle dynamics. The considered system parameters are: $R_g = 0.033$, $L_g = 0.33$, $\omega_1 = 1$ and $V_{set} = 1$.

Fig. 3 (left figure) depicts the sufficient region for the frequency and voltage droops (derived using (20)), that provide a good approximation of the RMS model. As it can be seen from the figure the impact of the voltage droop on the frequency droop is negligible. It should be noted that this occurs due to the high X/R ratio of the grid impedance. As it will be presented in Section 3.3, low X/R ratio increases the influence of the voltage droop on the maximum value of the frequency droop. Fig. 3 (right figure) shows the gain margin for different values of the control parameters k_v and K_q . Based on the proposed control design, the impact of the voltage droop on the gain margin is negligible compared to k_v .

Fig. 4 depicts the Bode magnitude plot of the transfer function $e(s)$. $e(s)$ describes the error between the RMS and EMT model. K_p is equal to the upper bound defined by (20), $K_q = 0$ and k_v is determined by (21a). As it can be seen in the figure, for small values of g_m , the magnitude of $e(s)$ is of significance around the nominal frequency of the system. This indicates that the power signal captured by the RMS model will present an important error with respect to the actual power signal (captured by the EMT model). Increasing the gain margin decreases the approximation error, as shown in the same figure. We can also notice that for $\omega \leq 0.1\omega_1$, the impact from increasing the gain margin and consequently k_v on the amplitude of $e(s)$ is limited. This indicates that the value of the frequency droop is the one that primarily affects the phase shift of the open loop system and consequently the time response of the active power controller. Moreover, it can be seen that the gain margin is a useful metric for analyzing the H_∞ norm of the $e(s)$. By selecting a high gain margin we can limit the error between the RMS and EMT not only

within the range of $[0, \omega_c]$ but also over the whole frequency spectrum. Lastly, as shown in the figure, the singular values $e(s)$ at steady state, $\omega = 0$, are equal to zero, which was expected since the RMS model is able to preserve the dc-gain of the EMT state-space model. It is also depicted that during a dynamic event $\omega > 0$, the RMS model presents a mismatch with respect to the EMT model.

To summarize the results, we show that by selecting a high gain margin leads to a good-approximation of the RMS model over the whole frequency spectrum, where, on the other hand, by limiting the gain crossover frequency, the RMS and EMT models have the same phase shift at the frequencies at which the plant is supposed to track its reference signal ($[0, \omega_c]$).

3.2. Virtual synchronous generator

Until now, we have neglected the filter constant and its impact on both the stability of the closed loop system and the time-scale separation between the frequency and the network line dynamics. In this section, we investigate its impact on the stability of the closed loop system, as well as on the selection of the frequency droop value. The main advantage of using a low-pass filter is that we can decouple the network line dynamics from the frequency dynamics by placing the cut-off frequency at $0.1\omega_1$. The advantage of placing a low-pass filter on the measurement is that it increases the damping of the line resonance. As a result, the active damping controller it could be disregarded, which implies that $k_v = 0$.

We first calculate the phase crossover frequency. To determine the gain crossover frequency, (15) is used with $C(s) = \frac{K_p}{s} \frac{\omega_f}{s + \omega_f}$. The Laplace complex variable s is set to $j\omega_{180}$ and $\Im(L(j\omega_{180})) = 0$ is solved. This yields

the expression in (22). We continue by determining the gain crossover frequency: the Laplace complex variable s is set to $j\omega_c$ and $|L(j\omega_c)| = 1$ is solved for ω_c . To realize the calculation of ω_c and obtain analytical expressions for the phase margin and gain parameters, we perform a Taylor series expansion of the transfer function describing the plant dynamics, $G_{\theta P}(s)$, which yields the following expression: $G_{\theta P}(s) = B - B_t s$. As indicated also in [11], this approximation is valid for $\omega \leq 0.1\omega_1$, i.e. the expression $G_{\theta P}(s) = B - B_t s$ is not valid for $\omega \gg 0.1\omega_1$. The mathematical expression for ω_c is given by (23).

$$\omega_{180} = \sqrt{\frac{K_q L_g \omega_1 + (L_g \omega_1)^2 + R_g^2}{L_g \left(2 \frac{R_g}{\omega_f} + L_g\right)}} \quad (22)$$

$$\omega_c = \frac{\sqrt{-\omega_f^2 (1 - (B_t K_p)^2) + \omega_f \sqrt{4(B K_p)^2 + \omega_f^2 (1 - (B_t K_p)^2)^2}}}{\sqrt{2}} \quad (23)$$

where

$$B_t = 2R_g L_g U_{\text{set}}^2 \frac{K_q + \omega_1 L_g}{(K_q \omega_1 L_g + (\omega_1 L_g)^2 + R_g^2)^2}$$

$$B = U_{\text{set}}^2 \frac{K_q \omega_1 L_g + \omega_1 L_g}{K_q \omega_1 L_g + (\omega_1 L_g)^2 + R_g^2}$$

In contrast to the power synchronization, where the value of the phase crossover frequency is larger than $\omega_{180} > \omega_1$, the phase crossover frequency of the virtual synchronous generator's open loop system can get values lower than ω_1 . From (22), we realize that when ω_f decreases the value of ω_{180} also decreases. To remove the effects of line resonance on the active power regulation we choose the gains of K_p , K_q and ω_f , so that the $\omega_c \leq 0.1\omega_{180}$. The motivation behind this constraint is that it ensures a minimum gain margin of 10, since the slope for $\omega_c \leq \omega_{180}$ is at least $-1/\text{decade}$ (-20dB/decade) due to the two poles at zero and $-\omega_f$ [24]; it also minimizes the impact of the pole corresponding to the network line dynamics on the phase shift, since the complex conjugate poles, associated with the network line dynamics, start influencing the phase of the open loop system at $\omega = \frac{\omega_n}{10\zeta}$ where ω_n is the natural frequency and ζ the damping ratio of the complex. This ensures an almost linear decrease of the phase until -180° . Using Mathematica, we extract the following conditions which satisfy $\omega_c \leq 0.1\omega_{180}$.

$$K_p \leq \frac{\omega_c}{\omega_f} \sqrt{\frac{\omega_c^2 + \omega_f^2}{(B^2 + B_t^2 \omega_c^2)}} \quad (24a)$$

$$\omega_c \leq 0.1\omega_1 \quad (24b)$$

$$\omega_f = \frac{\alpha_{\text{pm}} B \omega_c + B_t \omega_c^2}{B - \alpha_{\text{pm}} B_t \omega_c} \quad (24c)$$

$$\alpha_{\text{pm}} \geq \frac{\omega_c c_1}{c_2 + c_3 + c_4} \omega_1 L_g R_g \quad (24d)$$

$$\alpha_{\text{pm}} < \frac{K_q L_g + L_g^2 + R_g^2}{2L_g R_g \omega_c} \quad (24e)$$

where

$$c_1 = 0.66 (K_q L_g \omega_1 + R_g^2 + (L_g \omega_1)^2 (1 + \omega_c^2))$$

$$c_2 = \frac{1}{300} ((K_q L_g \omega_1 + R_g^2)^2)$$

$$c_3 = \frac{1}{300} (L_g^4 (1 - 100\omega_c^2) + (R_g L_g \omega_1)^2 (2 + 300\omega_c^2))$$

$$c_4 = \frac{1}{300} (K_q (L_g \omega_1)^3 (2 - 100\omega_c^2))$$

and the variable α_{pm} is the ratio $\frac{\Im(L(j\omega))}{\Re(L(j\omega))}$. Lastly, to evaluate the impact of the control and system parameters on the gain margin of the open loop system, we substitute ω_{180} into (14). This yields the following expression:

$$K_p = \frac{2R_g}{g_m (L_g^2 \omega_1)} \frac{((L_g \omega_1)^2 + R_g^2)((R_g + L_g \omega_f)^2 + (L_g \omega_1)^2)}{U_{\text{set}}^2 (2R_g + L_g \omega_f)^2 (K_q + \omega_1 L_g)} \quad (25)$$

$$+ \frac{2R_g K_q L_g \omega_1 (K_q L_g \omega_1 + (R_g + L_g \omega_f)^2 + R_g^2 + 2(L_g \omega_1)^2)}{U_{\text{set}}^2 (2R_g + L_g \omega_f)^2 (K_q + \omega_1 L_g)}$$

An analytical condition for the stability of the grid-forming converter and the maximum value of the frequency droop is obtained when $g_m = 1$. It should be noted that in case we want to further increase the damping of the line resonance and employ the active damping controller, the above expressions can be used for $\omega_v = 0$ and $R_g = R_g + k_v$. This means that the active damping controller would add as a pure resistance. In case, $\omega_v = 0.1\omega_1$, the conditions (23)-(25) remain the same, but the transfer function of the plant changes $G_{\theta P}(s) = B - B_t s$ (by the Taylor expansion). B and B_t are now different and are expressed as follows:

$$B_t = 2R_g L_g U_{\text{set}}^2 \frac{K_q + \omega_1 L_g}{(K_q \omega_1 L_g + (\omega_1 L_g)^2 + R_g^2)^2}$$

$$B = U_{\text{set}}^2 \frac{K_q \omega_1 L_g + \omega_1 L_g}{K_q \omega_1 L_g + (\omega_1 L_g)^2 + R_g^2}$$

It should be mentioned that as $\omega_v \rightarrow 0$, the approximation of the transfer function describing the plant dynamics becomes less accurate since $B_t \rightarrow \infty$. However, as mentioned above ω_v is usually chosen in the range $[0.1\omega_1, 0.2\omega_1]$.

3.2.1. Illustrative case

The following example will help us understand better the sufficient conditions (24a)-(24e). Similar to the previous case, the sufficient regions shown in Fig. 5 do not represent the stable region of the system, but the region that establish a good approximation of the RMS model of the grid-forming converter. Since in Section 3.1 we show that the

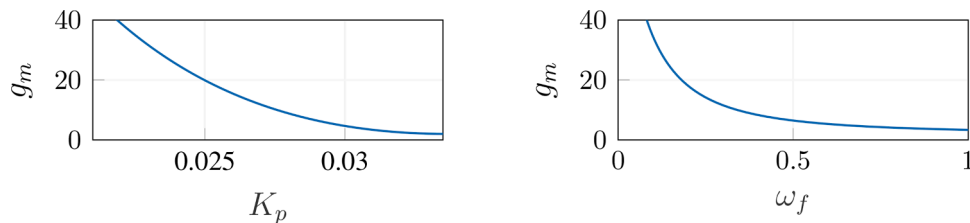


Fig. 6. Sufficient gain margin for different values of K_p (left) and ω_f (right).

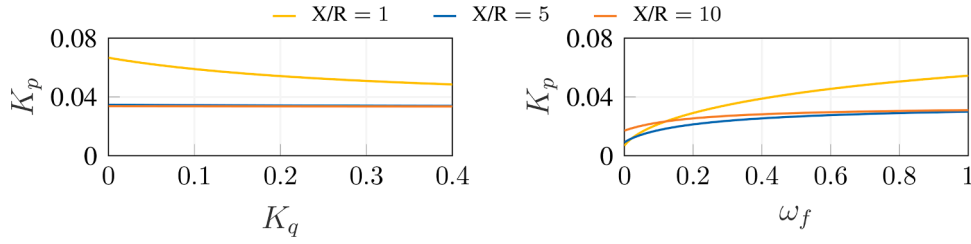


Fig. 7. Left: sufficient region for (K_q, K_p) for different X/R ratios. Right: sufficient region for (ω_f, K_p) for different X/R ratios.

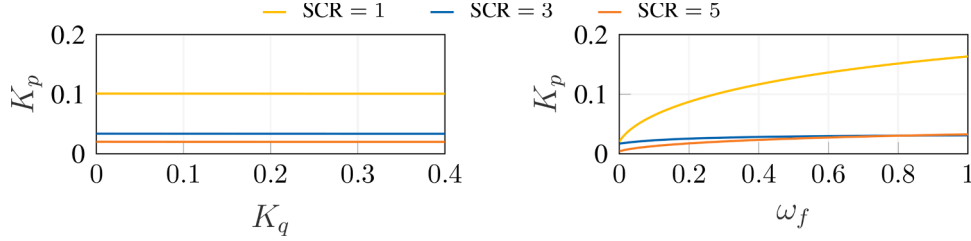


Fig. 8. Left: sufficient region for (K_q, K_p) for different values of SCR. Right: sufficient region for (ω_f, K_p) for different values of SCR.

voltage droop does not reduce the sufficient region defined by the control gains, we consider that $K_q = 0$. Fig. 5 shows the sufficient regions for the control parameters, the gain crossover frequency and the phase margin, extracted using (24a)-(24e). In the left figure, we show the sufficient region for the control parameters ω_f and K_p that enables the decoupling between the network line and frequency dynamics. We can see that by increasing ω_f , the maximum value of the frequency droop increases. In the right figure, we show the minimum value required for the phase margin, in order to mitigate the impact of network line dynamics on the active power controller. As ω_c approaches $0.1\omega_1$, the minimum required phase margin increases and so does the value of ω_f . Consequently, we can infer that the inertial effect of the grid-forming converter drastically reduces, when ω_c is close to $0.1\omega_1$.

Fig. 6 shows how the control and system parameters affect the gain margin of the open loop system and consequently the damping ratio of the mode associated with the line resonance. It can be seen that the gain margin of the open loop system remains larger than 2, if K_p and ω_f are chosen within the sufficient region defined by (24a)-(24e). Last but not least, it can be seen that as ω_c approach $0.1\omega_1$, and consequently $\omega_f \rightarrow 1$ the gain margin of the system decreases significantly. It should be noted that the value of the gain margin is significantly affected by the X/R ratio of the grid impedance. For lower values of the X/R ratio the gain margin is expected to be larger, however for larger resistances the influence on the phase of the open loop system becomes larger [11].

3.3. Transmission-line length & X/R ratio

We now investigate the impact of the transmission-line length and X/R ratio on the sufficient regions defined by the control parameters of the grid-forming converter. Fig. 7 shows how the different levels of X/R

ratio affect the sufficient regions for the control parameters. For the comparison, we consider that $L_g = 0.333$. We consider that, the X/R ratio is the ratio $\frac{L_g}{R_g}$. The left figure indicates that the lower the value of the X/R ratio the higher the impact of the voltage droop on the maximum value of the frequency droop. Moreover, it can be seen that the lower the X/R ratio the higher the maximum value for the frequency droop becomes. From the right figure, we observe that the maximum value of the frequency droop is similar for low values of the $\omega_f < 0.2$ and the different levels of X/R ratios. In the case of X/R=1, the maximum value for K_p increases, as $\omega_f \rightarrow 1$, which is not the case for higher levels of X/R.

Fig. 8 shows how different levels of Short Circuit Ratio (SCR) affect the sufficient regions for the control parameters. For the comparison, we consider that the X/R is equal to 10, which allows us to consider that $L_g \approx \frac{1}{SCR}$. The left figure indicates that the lower the value of the SCR, the higher the relative value of the frequency droop becomes. This shows that increase of transmission-line length extends the sufficient region, in which the quasi-static approximation of the network line dynamics is valid. We can also observe that the impact of the voltage droop on the frequency droop is negligible, since the maximum relative value for the frequency droop does not change. In the right figure, it is depicted the impact of the SCR on the sufficient region defined by the control gains K_p and ω_f . We can observe that the maximum relative value of the frequency droop increases for lower values of the SCR. Moreover, it can be seen that for low values of ω_f , the maximum value of the frequency droop is relatively small compared to the case where the low-pass filter is disregarded.

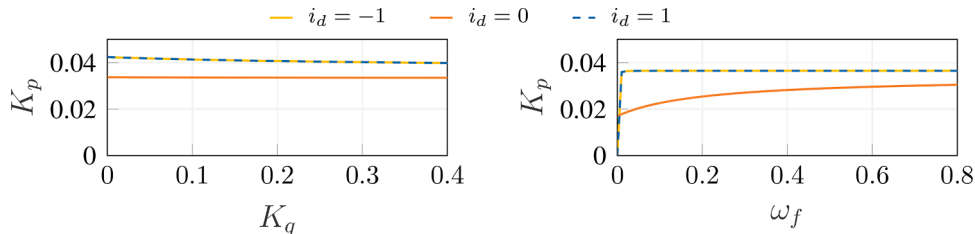


Fig. 9. Impact of active current on the sufficient region for the control gains of the power synchronization loop and virtual synchronous generator. For the virtual synchronous generator we consider that $K_q = 0$.

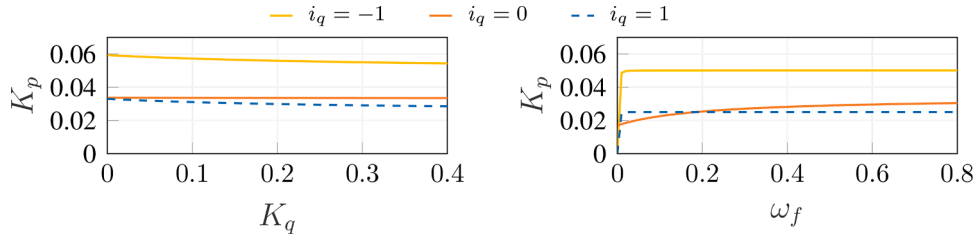


Fig. 10. Impact of reactive current on the sufficient region for the control gains of the power synchronization loop and virtual synchronous generator. For the virtual synchronous generator we consider that $K_q = 0$.

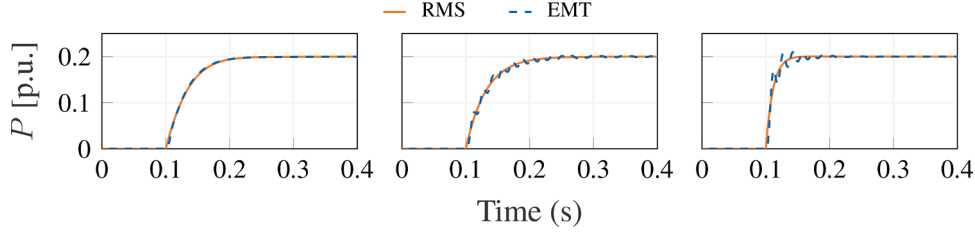


Fig. 11. Active power response of power synchronization loop. Left figure: Case 1, Middle figure: Case 2, Right figure: Case 3.

3.4. Impact of active and reactive currents

Fig. 9 and Fig. 10 demonstrate the impact of active and reactive currents, respectively, on the sufficient region defined by the control gains of the grid-forming converter. As it can be seen from the figures, the sufficient region slightly reduces, when the grid-forming converter absorbs reactive power. It should be noted that the increase of the sufficient region indicates that the bandwidth of the closed loop system of the active power controller decreases when the converter is initialized at different operating point. This means that in case of a small droop value, the active power controller might not be able to track the reference signal, which is something that was also highlighted in [22].

4. Simulation and results

In this section, we verify using time-domain simulations the sufficient regions extracted in the previous section. Our goal is to compare the active power response, obtained by the fully detailed model and the proposed reduced order models for different control and system parameters (see Table B1). For the analysis, we consider that the rated power of the converter is 1000 MVA.

4.1. Validation on 2-Bus system - Step change of the active power set-point

The presented sufficient conditions are evaluated on the 2 bus system depicted in Fig. 1. For the analysis we consider $L_g = 0.333$ and $R_g = 0.0333$. The control parameters for the inner loop control and the values of the phase reactor and filter capacitance can be found in the appendix.

4.1.1. Power synchronization loop

For the frequency droop-based control, we consider the following set

of parameters in order to evaluate the accuracy of the reduced order model which corresponds to the phasor-approximation model.

- Case 1: $K_p = .0335$, $K_q = 0.2$, $k_v = 0.12$
- Case 2: $K_p = .0335$, $K_q = 0.2$, $k_v = 0.0$
- Case 3: $K_p = .1$, $K_q = 0.2$, $k_v = 0.1$

For the good approximation case, the selection of the control gains is based on the conditions (20)-(21c). For the poor approximation case, we choose the frequency droop according to (20) and we neglect the active damping controller, which leads to reduced gain margin and poor damping of the line resonance. For the third case, we consider a high value of frequency droop which increases the bandwidth of the active power controller and a smaller value of damping resistor k_v .

Fig. 11 shows the converter's response for the step change of the P_{ref} and Fig. 12 the active power error between the RMS and EMT model for the 3 different cases. For the case where the frequency droop was selected based on the sufficient conditions, it can be seen that the difference between the RMS and EMT models is insignificant. For this case, the root mean squared error between the RMS and EMT model during the transient period is equal to 0.0036. Moreover, for the case where the active damping controller is neglected it can be seen that the impact of the line resonance on the system response increases, which was expected due to the poor gain margin of the open loop system. However, the time response of the RMS and EMT model matches, since the K_p is tuned according to (20). For this case, the relative mean square error between the RMS and EMT model during the transient period is equal to 0.0062. It can be seen that the active power signal is oscillatory with the frequency of oscillations being equal to ω_{180} (see (17)). In the case we use a high frequency droop value and a relative small value for the k_v , it can be seen that the maximum active power error between the RMS and EMT model is higher compared to the other cases, which indicates that the

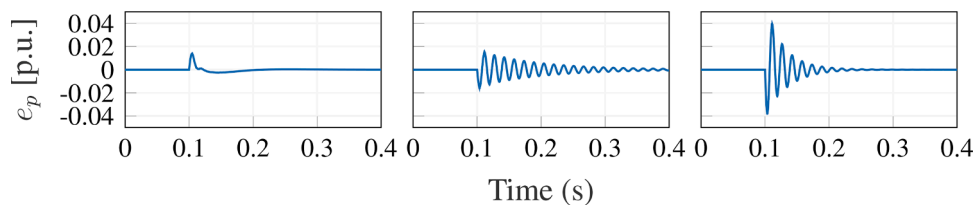


Fig. 12. Active power error between RMS and EMT - Power synchronization loop. Left figure: Case 1, Middle figure: Case 2, Right figure: Case 3.

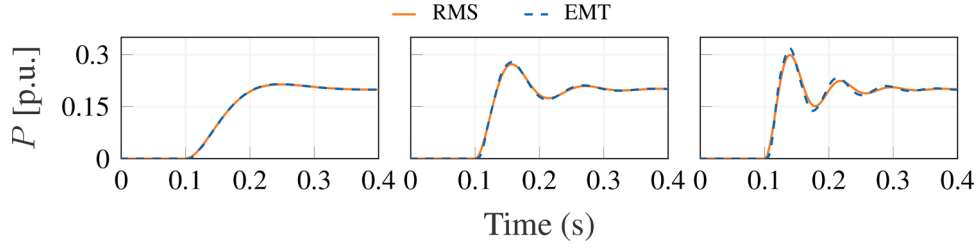


Fig. 13. Active power response of virtual synchronous generator. Left figure: Case 1, Middle figure: Case 2, Right figure: Case 3.

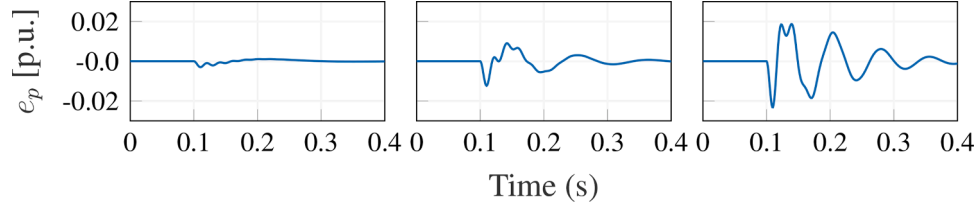


Fig. 14. Active power error between RMS and EMT - Virtual synchronous machine. Left figure: Case 1, Middle figure: Case 2, Right figure: Case 3.

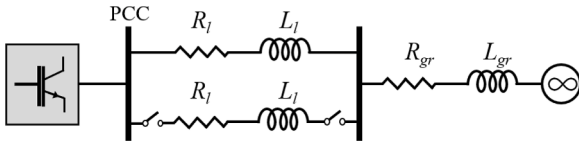


Fig. 15. Test system containing a single-line diagram of a converter connected to an infinite bus via two parallel transmission lines.

impact on the estimation error is higher when $\omega_c > 0.1\omega_1$. For this case, the relative mean square error between the RMS and EMT model during the transient period is equal to 0.0153.

4.1.2. Virtual synchronous generator

We consider the following set of parameters in order to evaluate the accuracy of the phasor approximation model. We consider three cases for the control parameters K_p and ω_f , where we gradually increase the frequency droop value, which consequently decreases the phase margin. For case 1, the control gains have been selected based on the sufficient conditions. For the considered cases, we consider that $K_q = 0$ and $k_v = 0$.

- Case 1: $K_p = .0230$, $\omega_f = 0.1108$
- Case 2: $K_p = .1$, $\omega_f = 0.1108$
- Case 3: $K_p = .2$, $\omega_f = 0.1108$

Fig. 13 demonstrates the active power response and Fig. 14 the active power error between RMS and EMT for the three different cases. Apart from the poor phase margin and the oscillatory response of the active power due to the decrease of the phase margin, it can be noticed that the maximum error between the RMS and EMT models is relatively higher for the 2nd and 3rd cases. This indicates that the phase margin predicted by the RMS model appear rather optimistic compared to the EMT model. Moreover, it can be seen that by increasing the frequency droop, the gain margin of the open loop system deteriorates, which leads to a higher influence of the line resonance on the angle dynamics. Lastly, compared to the power synchronization loop, we can observe that the error between the RMS and EMT is similar for the 1st and 2nd cases, where it is relatively higher for the 3rd case.

4.2. Validation on 2-Bus system - Line outage

To further validate the conditions, we consider a line outage in the system depicted in Fig. 15, which is simulated as a step decrease of the

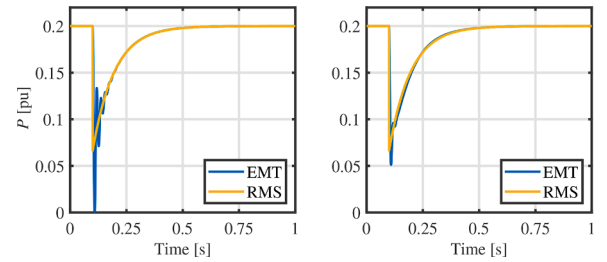


Fig. 16. Trajectory of the active power signal for a sudden drop of the SCR from 3 to 1 - Power synchronization loop. In the left figure the value of $k_v = 0.1333$ and in the right $k_v = 0.4$.

SCR value, while the X/R remains the same.

4.2.1. Power synchronization loop

Fig. 16 shows the active power signal for a change of the SCR from 3 to 1 in the case of power-synchronization loop control scheme. We want to show the effect of the damping resistor control gain k_v on the mismatch between the RMS and EMT modeling of the converter. In Fig. 16, we show the trajectory of the active power signal. In the left figure, k_v is equal to 0.1333 and in the right figure k_v is equal to 0.4. As it can be seen, the mismatch between the RMS and EMT model is higher in the left figure where the value of k_v is smaller and the RMS model of the converter is inadequate of capturing the maximum active power deviation. This comes to show the effect of damping ratio of the mode associated with the line dynamics. In particular, using the expression $g_{m,\min} = (k_v + R_g) \frac{20}{L_g}$, where $R_g = 0.0333$ and $L_g = 0.333$ at pre-fault state and $R_g = 0.1$ and $L_g = 1$ at post-fault, we can calculate the minimum gain margin of the open loop system $G_{ol}(s)$. For $k_v = 0.133$, although the gain margin is sufficiently large ($g_{m,\min} = 10$) at pre-fault state, it decreases significantly $g_{m,\min} = 4.66$ due to the event. For $k_v = 0.4$, the value of the gain margin is higher than 10 both at pre- and post-faults, which improves significantly the damping of the line resonance and therefore minimizes the error between the RMS and EMT model. Moreover, it can be seen that higher value of k_v limits the maximum active power deviation and consequently improves the dynamic performance of the system. It should be mentioned, however, that large values of k_v can decrease the bandwidth of the active power control and also deteriorate the phase margin of the open loop system [22], which implies that there is a trade-off between transient performance and

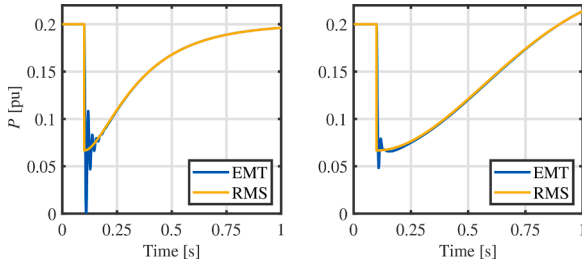


Fig. 17. Trajectory of the active power signal for a sudden drop of the SCR from 3 to 1 - Virtual synchronous generator. In the left figure the value of $k_v = 0.1333$ and in the right $k_v = 0.4$.

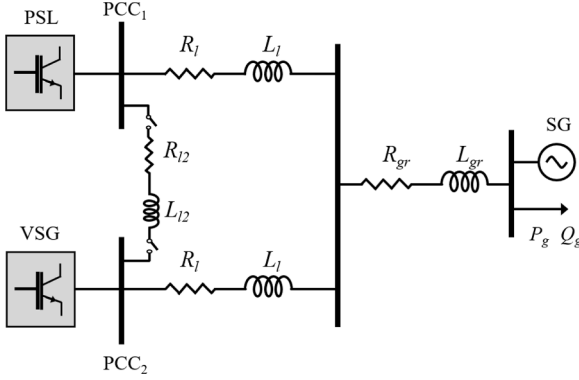


Fig. 18. Test system containing a single-line diagram of two grid-forming converters, one using Power Synchronization Loop (PSL) active power control and the other the Virtual Synchronous Generator (VSG). The grid-forming converters are connected to an equivalent variable frequency AC grid and a load.

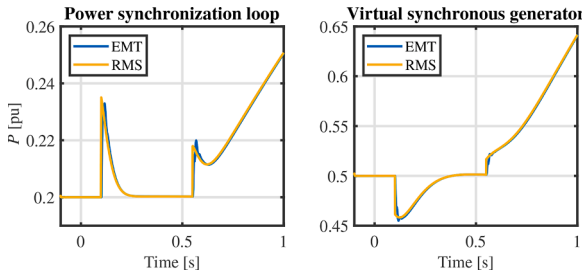


Fig. 19. Trajectory of the active power signal of the two converters.

maximum bandwidth of the active power controller of the grid-forming converter.

4.2.2. Virtual synchronous generator

Similar observations can be made in the case of the virtual synchronous generator control scheme, see Fig. 17. As it can be seen the mismatch between the RMS and EMT model is higher in case of a smaller k_v value. Also it can be noticed from the left figure that the settling time of the active power signal increases. This is due to the fact that higher value of k_v decreases the phase margin of the open loop system and consequently the damping ratio corresponding to the modes associated with the active power controller deteriorates.

4.3. Multi-converter topology

To validate if the conditions are valid in case of nearby connections of grid-forming converters, we consider the grid topology depicted in Fig. 18. Two events are realized: (i) line outage at $t=0.1$ s and (ii)

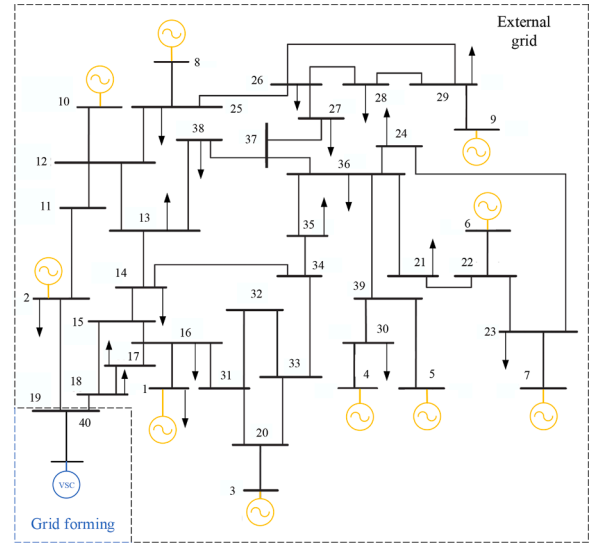


Fig. 20. Modified version of IEEE 39 Bus system. Disturbance location is indicated by red symbol. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

generation loss equal to 1000 MW at $t=0.55$ s. Before the line outage, the output impedance for both of the converters is equal to $L_g = 0.415$ and $R_g = 0.0366$ and after the line outage is equal to $L_g = 0.48$ and $R_g = 0.0453$. Both converters are of identical rated power. The converter utilizing the power synchronization loop operates with an active power equal to 0.2 pu at pre-fault state and the virtual synchronous generator with an active power equal to 0.5 pu. The control settings of the active power control loop for the converters are as follows:

- PSL: $K_p = 0.0417$, $K_q = 0.2$ and $k_v = 0.197$
- VSG: $K_p = 0.0135$, $\omega_f = 0.074$, $K_q = 0.2$ and $k_v = 0.197$

From Fig. 19, it can be concluded that the mismatch between the RMS and EMT models of the system is negligible (a bit higher in the case of power synchronization loop) when considering events that may occur in the transmission grid, as long as the control parameters are selected based on the conditions presented in Section 3 of the paper.

4.4. Validation on large system

The presented sufficient conditions are evaluated on a modified version of the IEEE 39 bus system, which is depicted in Fig. 20. The system parameters, the control gains and the components comprising the test case can be found in [26]. A grid-forming converter is placed on bus 40. For the analysis and the validation of the phasor approximation model, we simulate a step change of P_{ref} from 0 to 200 MW. The simulations have been performed in PowerFactory, using the RMS and EMT simulation methods [27].

4.4.1. Power synchronization loop

Fig. 21 depicts the active power response for the 3 different cases. Similar to the previous case and the 2 bus system, it is evident that the active damping controller mitigates the effect of the line resonance on the angle dynamics. Consequently, the mismatch between the RMS and EMT simulation methods is insignificant. As in the 2 bus system, for high values of frequency droop and smaller value of k_v , the mismatch between the RMS and EMT model of the converter is significantly larger.

4.4.2. Virtual synchronous generator

For the analysis three cases have been considered; the control parameters ω_f and K_p are the ones that are used for the one bus system

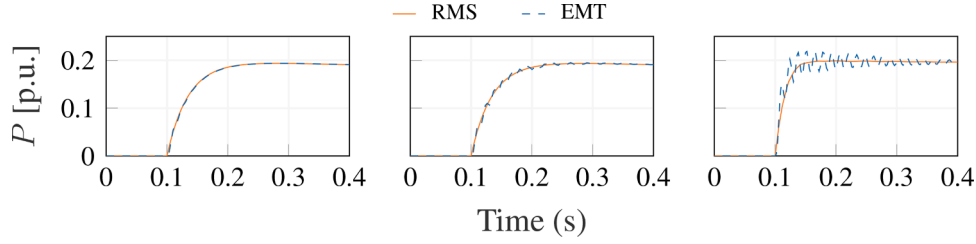


Fig. 21. Active power response of power synchronization loop - Powerfactory. Left figure: Case 1, Middle figure: Case 2, Right figure: Case 3.

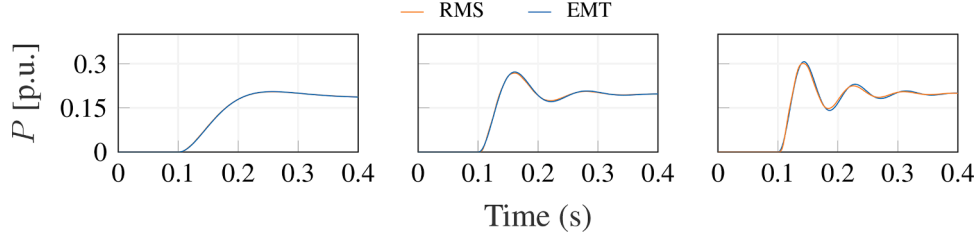


Fig. 22. Active power response of virtual synchronous generator - Powerfactory. Left figure: Case 1, Middle figure: Case 2, Right figure: Case 3.

configuration. Fig. 22 depicts the active power response of the synchronous generator. As it can be seen the higher the frequency droop, the smaller becomes the phase margin of the open loop system, which leads to high low frequency oscillations, as well as decrease of the gain margin of the open loop system. Similar to the previous case and the 2 bus system, we can observe a mismatch between the RMS and EMT models in the maximum value of the active power signal, as well as a small phase shift between the two responses.

5. Conclusion

In this paper, we study the appropriateness of phasor-approximation models when simulating grid-forming converters. We present sufficient conditions for the control gains of outer loops of the grid-forming converter, that enable transmission system operators to use the RMS model of the grid-forming converter for monitoring its active power exchange with the grid. We also provide sufficient bounds for the control gains that ensure the stability of the full-detailed model. We provide expressions that can be used for sufficient damping of the line resonance and mitigation of the impact of network line dynamics on angle dynamics. Lastly, we demonstrate that in the case of the power synchronization loop, the RMS model is unable to detect the oscillations at the fundamental frequency for relative small value of the frequency droop, as long

as the active damping controller is neglected. On the contrary, for the virtual synchronous generator, the RMS and EMT models can differ on the phase response (depending on the ratio of the damping and inertia parameter), as well as on the maximum active power deviation caused by the poor gain margin of the open loop system.

Future work is intended to investigate the impact of a slower voltage droop controller (low pass filter on the reactive power), high values of SCR and different classes of grid-forming converters on the appropriateness of RMS modeling.

CRediT authorship contribution statement

George S. Misyris: Conceptualization, Methodology, Software, Investigation, Validation, Writing - original draft. **Spyros Chatzivasilaidis:** Writing - review & editing. **Tilman Weckesser:** Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Derivation of the transfer function describing the plant dynamics

We derive the transfer function describing the relation between $\Delta\theta$ and Δi , while accounting for the impact of voltage droop on the active power regulation. We consider that $U = U_{set} + \Delta U$ and $U_s^c = U_s$ at steady state. Using perturbation quantities of θ , U and i , (4) is expressed as:

$$U_{set} + \Delta U - (R + [s + j(\omega_1 + \dot{\Delta}\theta)]L_g)(i_0 + \Delta i) = U_s e^{-j(\theta_0 + \Delta\theta)}. \quad (A.1)$$

where, $\Delta U = -H_{hp}(s)\Delta i + K_q\Delta i_q$ if neglect the term $\Delta U\Delta i_q = 0$. The term $e^{-j\Delta\theta} \approx 1 - j\Delta\theta$. The approximation stands for small variation of $\Delta\theta \approx 0$, since $\cos(\Delta\theta) \approx 1$ and $\sin(\Delta\theta) \approx \Delta\theta$. This can be explained by:

$$e^{-j\Delta\theta} = \underbrace{\cos(\Delta\theta)}_1 - \underbrace{j\sin(\Delta\theta)}_{j\Delta\theta}. \quad (A.2)$$

After these simplifications (A.1) is expressed as:

$$-K_q\Delta i_q + [R_g + H_{hp}(s) + \underbrace{(s + j\omega_1)L_g \Delta i - j(U_s e^{-j\theta_0} - sLi_0)\Delta\theta + U_{set} - (R_g + j\omega_1 L_g)i_0 - U_s e^{-j\theta_0}}_{=0}] \Delta i = 0 \quad (A.3)$$

where $U_{\text{set}} - (R_g + j\omega_1 L_g)i_{\theta} - U_s e^{-j\theta_0} = 0$ (steady state condition). Using (A.3), we can derive the relation between $\Delta\theta$ and Δi :

$$\underbrace{\begin{bmatrix} H_{\text{hp}}(s) + R_g + L_g s - K_q - \omega_1 L_g \\ \omega_1 L_g H_{\text{hp}}(s) + R_g + L_g s \end{bmatrix}}_A \Delta i = \underbrace{\begin{bmatrix} \omega_1 L_g i_{\theta 0} + i_{\theta 0} (R_g + L_g s) \\ \omega_1 L_g i_{\theta 0} - i_{\theta 0} (R_g + L_g s) + U_{\text{set}} \end{bmatrix}}_B \Delta \theta. \quad (\text{A.4})$$

where $A^{-1}B$ provides the relationship between $\Delta\theta$ and $\Delta i_{d,q}$ and is a vector of two rows and one column. We consider that $G_{\theta i_d}(s) = \frac{\Delta i_d}{\Delta \theta}$, $G_{\theta i_q}(s) = \frac{\Delta i_q}{\Delta \theta}$ and $G_{\theta \Delta i}(s) = [G_{\theta i_d}(s) \ G_{\theta i_q}(s)]^T$. Similar to [22], by using perturbation variables we can express the active power variation of the VSC as:

$$\Delta P = \Re(U_{\text{set}} \Delta i^* + i_{\theta 0}^* \Delta U). \quad (\text{A.5})$$

Substituting (A.4) into (A.5) and considering that $\Delta U = -H_{\text{HP}}(s)\Delta i + K_q \Delta i_q$ yields:

$$\Delta P = \underbrace{\Re(U_{\text{set}} G_{\theta \Delta i}^*(s) - i_{\theta 0}^* H_{\text{HP}}(s) G_{\theta \Delta i}(s))}_{G_{\text{HP}}(s)} + i_{\theta 0}^* K_q G_{\theta i_q}(s) \Delta \theta. \quad (\text{A.6})$$

Lastly, to extract the transfer function of the plant dynamics corresponding to the RMS model, we set the Laplace operator s in (A.3) equal to zero. We do that in order to extract the transfer function $e(s)$ which represents the difference between the RMS and EMT model of the active power controller.

Appendix B. Grid-forming converter parameters

Table B1
Converter parameters.

Parameter	Value	Parameter	Value
Rated power	1000 MVA	filter inductance	0.15 pu
Rated AC voltage	320 kV	filter capacitance	0.066 pu
Rated DC voltage	640 kV	filter resistance	0.005 pu
Switching frequency	5 kHz	Current controller proportional gain	2.3873
Current controller integral gain	25	Voltage controller proportional gain	0.59
Voltage controller integral gain	736	Voltage/current feed-forward terms	1

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